

Mat Lowery

Software Engineer · Denver, CO

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ABOUT

I build internal platforms and infrastructure for engineers who have better things to worry about. A decade at eBay's scale — Kubernetes control planes, etcd, DNS, container registries, federation — taught me that the gap between 'it works' and 'it's easy to operate' is where most of the real work lives.

IMPACT

- **\$200K+ Annual vendor cost eliminated:** Led container registry migration from proprietary to open-source infrastructure.
- **75% DNS query volume reduction:** Tuned CoreDNS behavior and contributed upstream improvements.
- **50% Pager volume reduction:** Prioritized SLO signals and follow-up fixes with the API Server SRE team.
- **2x+ Container image upload rate:** Patched container registry upload code and tuned chunk size for faster pushes.

SKILLS

- **Languages:** Go, Python, Shell
- **Platform and Kubernetes:** Docker, Kubernetes, API Server, CoreDNS, DNS, OpenStack, Operators, Federation, OCI container registries, Quay, API Priority & Fairness
- **Infrastructure and delivery:** GitOps, Prometheus, Grafana, Tekton, Linux, Release automation, Disaster recovery
- **Databases:** MySQL, etcd, Redis
- **Tools:** Jira, Confluence, PagerDuty
- **Practices:** Mentoring, Technical writing
- **AI-assisted engineering:** Codex, Gemini, Claude

WORK EXPERIENCE

eBay — Senior MTS, Software Engineer (Senior Staff)

Feb 2022 – Apr 2026

Owned and operated Kubernetes control planes, etcd, DNS, federation, container registry, and platform automation at one of the industry's largest private Kubernetes fleets • Led a container registry migration that eliminated six figures per year in vendor cost • Patched container registry upload code and tuned chunk size for 2x+ faster pushes • Improved API Server reliability with SRE partners, reducing pager volume 50% • Rolled out operator-based automation across 100 etcd clusters • Sharded federation syncing for multi-year scale runway

eBay — MTS 2, Software Engineer (Staff)

May 2016 – Feb 2022

Built and operated eBay's Kubernetes platform including the control planes themselves on eBay compute and additionally cluster DNS, cluster federation, release automation, security hardening, and production support at 60+ clusters, 30K+ nodes, and 160K+ pods • Tuned CoreDNS to reduce query volume 75%, increase QPS 4x, and cut CPU and memory by 80% and 50% • Built GitOps-based federation release and credential rotation automation

eBay — MTS 2, Software Engineer (Staff)

Nov 2013 – Apr 2016

Developed database-as-a-service for MySQL, MongoDB, Couchbase, and Cassandra on eBay infrastructure • Hardened database failover tooling by raising test coverage from 0% to 75% with minimal core-code changes • Contributed code, reviews, bugs, documentation, and infrastructure debugging across OpenStack Trove and related OpenStack community projects

X.commerce — MTS 1, Software Engineer (Senior)

Apr 2012 – Nov 2013

Built and operated private OpenStack infrastructure on bare metal, including Nova, Keystone, Glance, Swift, and Cinder • Led replacement of floating-IP failover with Corosync and Pacemaker clustering for more robust VM high availability

Pentaho — Senior Software Engineer

Oct 2006 – Apr 2012

Built core backend platform features for business intelligence products, including secured artifact repository architecture, security infrastructure, and open-source integrations • Led product security components and wrote public documentation used by customer-facing teams • Contributed to Spring Security, Apache Jackrabbit, and CloudInit

Wyndham Worldwide — Application Software Developer

Oct 2003 – Sep 2006

Built enterprise web application infrastructure for customer-facing systems, with focus on security, shared services, reusable application layers, and user-management services across Active Directory and Oracle user stores

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EDUCATION

University of Central Florida — Master's Degree, Computer Science 2001 – 2003

University of Central Florida — Bachelor's Degree, Computer Science 1998 – 2001

OPEN SOURCE

CoreDNS — [CoreDNS Kubernetes plugin namespace performance](#) : Merged a Kubernetes plugin change that replaced namespace list scans with cache lookups and reduced namespace cache memory for clusters with 10K+ namespaces.

Gophercloud — [Gophercloud reauth deadlock analysis](#) : Reported and isolated a reauthentication deadlock with a minimal reproducer and lock-level diagnosis.

Quay — [RadosGW storage provider chunk sizing](#) : Opened a Quay storage provider change adding `minimum\_chunk\_size\_mb` configurability for RadosGW uploads.

Kubernetes — [Kubelet pod status reason/message bug](#) : Filed a SIG Node bug showing how kubelet could copy `status.reason` and `status.message` across pod phase transitions.

OpenStack Trove — [OpenStack Trove cluster task manager](#) : Merged the cluster task manager implementation for OpenStack Trove, adding the orchestration layer for database cluster lifecycle work.

OpenStack Trove — [Service endpoints from catalog](#) : Merged a Trove service-catalog change that moved service endpoint discovery away from hard-coded URL configuration.

PERSONAL PROJECTS

emcee — Multi-cluster command runner for running commands in parallel across Kubernetes clusters.

sticky-kubeconfig — Shell helper for keeping a unique kubeconfig per terminal session.

executable-image — Shell tooling to run Docker images as executables.

kubectl-watchhook — kubectl plugin to watch Kubernetes objects and call a command per watch event.